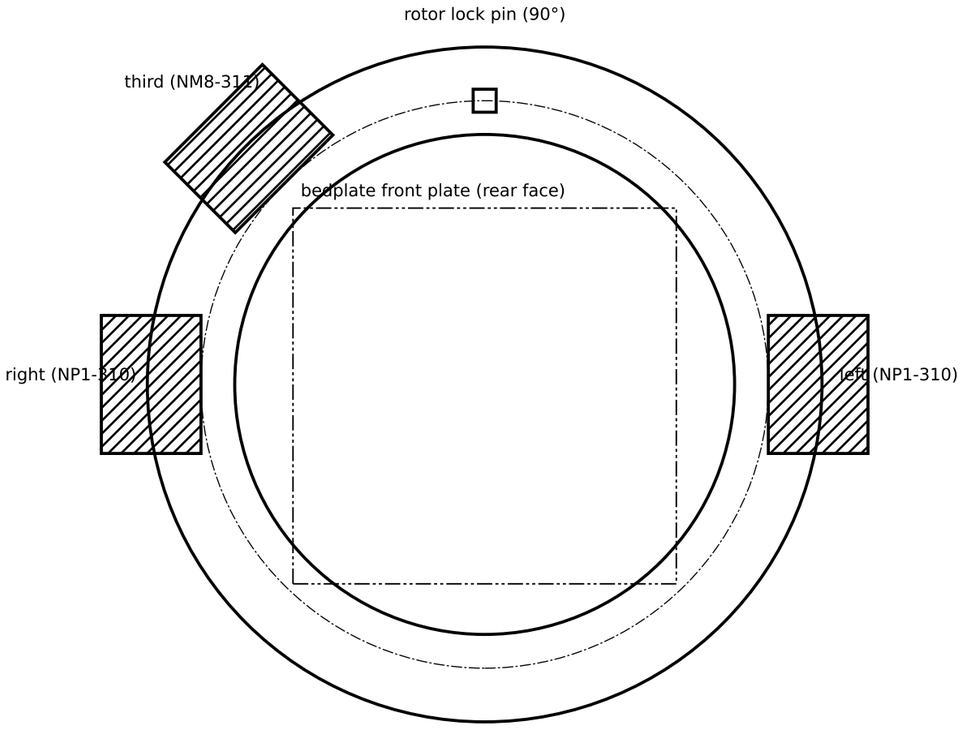
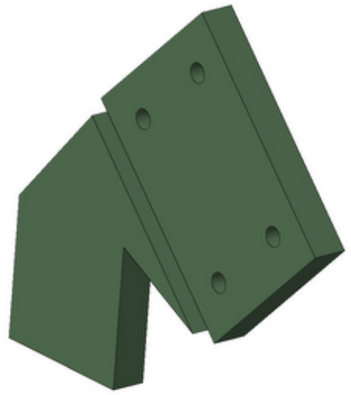
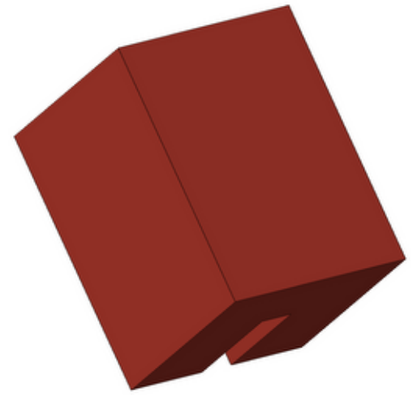
VIEW FROM THE REAR (downwind), angles from +y towards +z



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Brake (spindle_m8 / hub_yaw_m8)	Value
Required holding torque	14.4 kNm (parked aero torque at 8 m, design; 18 m: 13.1)
Calipers	3 × spring-applied, 4.8 kNm each
Clamp force per caliper	16.2 kN at μ 0.40, r_eff 0.37 m
Parked aero torque (sim, char.)	9.32 kNm at 18 m, 10.59 kNm at 8 m
Disc	NP1-230 Rev B, 16 mm, unchanged

NOTES

1. The third caliper sits at 135° (upper right seen from the rear) because the rotor lock pin holder occupies 90°. It uses the same caliper and mounting plate as NP1-310 (4 × M16 axial).
2. Bracket: 25 mm mounting plate + 25 mm web + 25 mm foot, bolted 4 × M16 10.9 through the bedplate front plate outside the spindle flange bolts (r ≥ 205). Weld a5, MT 100 %.
3. Alternative (decision D2 of doc 35): two calipers with a stronger spring pack (7.2 kNm each) in the existing positions; the third bracket then goes.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

ROTOR BRAKE · THREE CALIPERS AND THIRD CALIPER BRACKET

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-311	A · 2026-11-06	1:8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
bracket S355J2+N 25 plate, welded	bracket 7.0 kg	Claude / not checked